Multiple Imputation

This is the continuation of the experiment reports. The focus of this notebook is multiple imputation in the Titanic dataset.

This notebook includes the missing values treatment and imputations as well as an exploratory analysis of the imputed datasets. Finally, it includes a baseline model for the Titanic challenge.

Library Downloads

Copy

```
library(Amelia)
library(mice)
library(Hmisc)
library(missForest)
library(randomForest)
library(VIM)
library(tidyverse)
```

Sample Description

First, the first sample is loaded.

Copy

```
# -----------
# Loading the sample
# -----------
cat("\n Reading from the Average Transaction Filling/Imputing (Fill/Impute) Sample\n\n")
set.seed(seed=1234)

load_data <- read.table("/kaggle/input/Titanic-Fill-Impute/train.csv",
                        header=TRUE,
                        sep=",")
cat("\n\n Dimensions of the data\n\n")

dim(load_data)

str(load_data)
# -----------
# Reading from the Average Transaction Filling/Imputing (Fill/Impute) Sample
# -----------

Dimensions of the data

dim: int [1:2] 891 12

'data.frame':   891 obs. of  12 variables:
 $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Survived   : int  0 1 1 1 0 0 0 0 1 1 ...
 $ Pclass     : int  3 1 3 1 3 3 1 3 3 2 ...
 $ Name       : chr  "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)" "Heikkinen, Miss. Laina" "Futrelle, Mrs. Jacques Heath (Lily May Peel)" ...
 $ Sex        : chr  "male" "female" "female" "female" ...
 $ Age        : num  22 38 26 35 35 NA 54 2 27 14 ...
 $ SibSp      : int  1 1 0 1 0 0 0 3 0 1 ...
 $ Parch      : int  0 0 0 0 0 0 0 1 2 0 ...
 $ Ticket     : chr  "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
 $ Fare       : num  7.25 71.28 7.92 53.1 8.05 8.46 51.86 21.075 11.133 30.071 ...
 $ Cabin      : chr  "" "C85" "" "C123" ...
 $ Embarked   : chr  "S" "C" "S" "S" ...

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 $ Cabin      : chr  "" "C85" "" "C123" ...
 $ Embarked   : chr  "S" "C" "S" "S" ...
```

Copy and paste the first sample into the variable "dataset" and remove the variables not relevant for analysis (PassengerID, Name, Ticket, and Cabin). In addition, the total absence of NA values within the answers is calculated.

Copy

```
# -----------
# Copy the data into a variable called "dataset"
# -----------
dataset <- load_data

# -----------
# Removing the variables that are not relevant for analysis (Passenger ID, Name, Ticket, and Cabin)
# -----------
dataset <- dataset[,-c(1,9,10,11)]

# -----------
# Calculating the total number of NA values in the dataset
# -----------
cat("\n\n Total number of NA values in the dataset: ",
    sum(is.na(dataset)),
    "\n\n")

# -----------
# Calculating the total number of NA values in the dataset
# -----------

Total number of NA values in the dataset:  687
```

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# -----------
# Calculating the total number of NA values in the dataset
# -----------

Total number of NA values in the dataset:  687
```

Copy

```
# -----------
# Calculating the number of NA values in each variable
# -----------
total_missing_data <- colSums(is.na(dataset))

cat("\n\n Number of NA values in each variable\n\n", total_missing_data)

# -----------
# Calculating the number of NA values in each variable
# -----------

 Number of NA values in each variable

 Survived   Pclass      Sex      Age    SibSp    Parch     Fare Embarked
        0        0        0      177        0        0        0        2
```

Copy

```
# -----------
# Calculating the percentage of NA values in each variable
# -----------
percentage_missing_data <- round(total_missing_data/dim(dataset)[1]*100,2)

cat("\n\n Percentage of NA values in each variable\n\n", percentage_missing_data)

# -----------
# Calculating the percentage of NA values in each variable
# -----------

 Percentage of NA values in each variable

 Survived   Pclass      Sex      Age    SibSp    Parch     Fare Embarked
     0.00     0.00     0.00    19.87     0.00     0.00     0.00     0.22
```

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```

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DEF975_DEF
DEF976_DEF
DEF977_DEF
DEF978_DEF
DEF979_DEF
DEF980_DEF
DEF981_DEF
DEF982_DEF
DEF983_DEF
DEF984_DEF
DEF985_DEF
DEF986_DEF
DEF987_DEF
DEF988_DEF
DEF989_DEF
DEF990_DEF
DEF991_DEF
DEF992_DEF
DEF993_DEF
DEF994_DEF
DEF995_DEF
DEF996_DEF
DEF997_DEF
DEF998_DEF
DEF999_DEF
DEF1000_DEF
```

Here, that the problem I want to talk, so, so,towards the first actual mouthpiece. To ensure,basically, a, and so, the Chinese, that, and so,and so,and so,

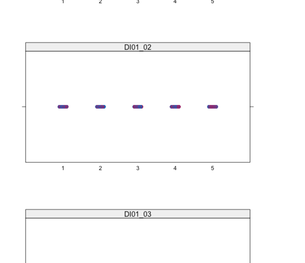
```
gtsim_dictionaryset_dictionaryset = gtsim_dictionaryset.keys()
for key in gtsim_dictionaryset_dictionaryset:
    print("gtsim_dictionaryset_dictionaryset[",key,"] = ",gtsim_dictionaryset_dictionaryset[key], sep = "")
```

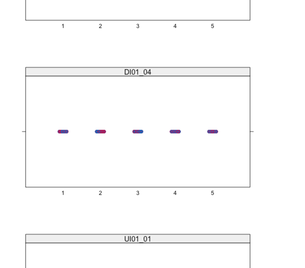
```
gtsim_dictionaryset_dictionaryset = gtsim_dictionaryset.keys()
for key in gtsim_dictionaryset_dictionaryset:
    print("gtsim_dictionaryset_dictionaryset[",key,"] = ",gtsim_dictionaryset_dictionaryset[key], sep = "")
```

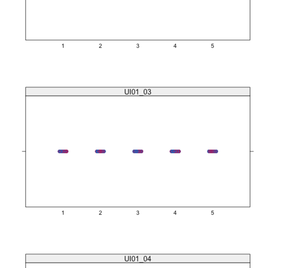
```
gtsim_dictionaryset_dictionaryset[ Sex ] = [ 'M' ]
gtsim_dictionaryset_dictionaryset[ Age ] = [ '18' ]
gtsim_dictionaryset_dictionaryset[ Pclass ] = [ '1' ]
gtsim_dictionaryset_dictionaryset[ Embarked ] = [ 'S' ]
gtsim_dictionaryset_dictionaryset[ Survived ] = [ '0' ]
gtsim_dictionaryset_dictionaryset[ Name ] = [ 'Mr. Owen Harris Braund' ]
gtsim_dictionaryset_dictionaryset[ Ticket ] = [ 'A/5 21171' ]
gtsim_dictionaryset_dictionaryset[ Fare ] = [ '7.2500' ]
gtsim_dictionaryset_dictionaryset[ Cabin ] = [ '#NULL!' ]
gtsim_dictionaryset_dictionaryset[ SibSp ] = [ '1' ]
gtsim_dictionaryset_dictionaryset[ Parch ] = [ '0' ]
gtsim_dictionaryset_dictionaryset[ PassengerId ] = [ '1' ]
gtsim_dictionaryset_dictionaryset[ Survival_Probability ] = [ '0.0000000000' ]
gtsim_dictionaryset_dictionaryset[ Survival_Prediction ] = [ '0' ]
gtsim_dictionaryset_dictionaryset[ Survival_Accuracy ] = [ '0.0000000000' ]
gtsim_dictionaryset_dictionaryset[ Survival_Error ] = [ '0.0000000000' ]
```

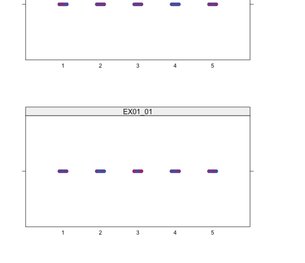
The attached print containing the data source is transcribed back into a data sheet for analysing the Titt Wilkes Lord and Whitehead data set. The data set is used for the purpose of the study of survival of the passengers on the Titanic. The data set is used for the purpose of the study of survival of the passengers on the Titanic. The data set is used for the purpose of the study of survival of the passengers on the Titanic

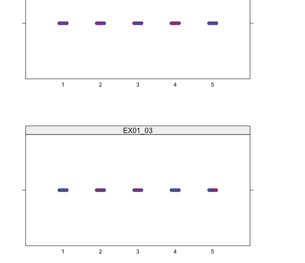
```
[ 1.5E-02, 1.5E-02, 1.5E-02, 1.5E-02, 1.5E-02, 1.5E-02,
  
```

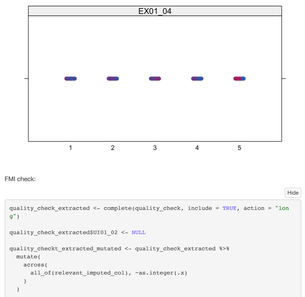


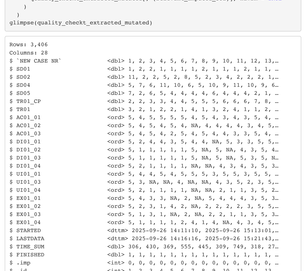


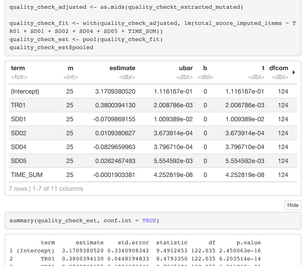












FMI check:

Hide

```
quality_check_extracted <- complete(quality_check, include = TRUE, action = "lon
g")

quality_check_extracted$UI01_02 <- NULL

quality_checkt_extracted_mutated <- quality_check_extracted %>%
  mutate(
    across(
      all_of(relevant_imputed_col), ~as.integer(.x)
    )
  )

quality_checkt_extracted_mutated <- quality_check_extracted %>%
  mutate(
    total_score_imputed_items =
    rowMeans(
      quality_checkt_extracted_mutated[, (relevant_imputed_col)], na.rm = TRUE
    )
  )
glimpse(quality_checkt_extracted_mutated)
```

```
Rows: 3,406
Columns: 28
$ `NEW CASE NR`            <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,…
$ SD01                     <dbl> 1, 2, 2, 1, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, …
$ SD02                     <dbl> 11, 2, 2, 5, 2, 8, 5, 2, 3, 4, 2, 2, 2, 1,…
$ SD04                     <dbl> 5, 7, 6, 11, 10, 6, 5, 10, 9, 11, 10, 9, 6…
$ SD05                     <dbl> 7, 2, 6, 5, 4, 4, 4, 4, 6, 4, 4, 4, 2, 1, …
$ TR01_CP                  <dbl> 2, 2, 3, 3, 4, 4, 5, 5, 5, 6, 6, 7, 8, …
$ TR01                     <dbl> 3, 2, 1, 2, 2, 2, 1, 4, 1, 3, 2, 4, 1, 1, 2, …
$ ACH1_01                  <ord> 5, 4, 5, 5, 5, 5, 4, 5, 5, 4, 3, 4, 5, 4, …
$ ACH1_02                  <ord> 5, 4, 5, 4, 5, 4, 3NA, 4, 4, 4, 4, 4, 3, 4, 5,…
$ ACH1_03                  <ord> 5, 4, 5, 4, 2, 5, 4, 5, 4, 4, 3, 3, 5, 4, …
$ DI01_01                  <ord> 5, 2, 4, 4, 3, 5, 4, 4, 4, NA, 5, 3, 3, 5, 5,…
$ DI01_02                  <ord> 5, 1, 1, 1, 1, 1, 5, NA, 5, NA, 4, 3, 5, 4,
$ DI01_03                  <ord> 5, 2, 1, 1, 1, 1, 5, NA, 5, NA, 5, 3, 5, 4,
$ DI01_04                  <ord> 5, 2, 1, 1, 1, 1, NA, NA, 4, 3, 4, 3, 5, 3,
$ UI01_01                  <ord> 5, 4, 4, 5, 4, 4, 5, 5, 5, 5, 3, 5, 5, 5, …
$ UI01_03                  <ord> 5, 3, NA, NA, 4, NA, NA, 4, 3, 5, 2, 3, 5,…
$ UI01_04                  <ord> 5, 2, 1, 1, 1, 1, NA, NA, 2, 1, 1, 3, 5, 2,
$ EX01_01                  <ord> 5, 4, 3, 3, NA, 2, NA, 5, 4, 4, 4, 3, 5, 3,
$ EX01_02                  <ord> 5, 2, 3, 1, 4, 2, NA, 2, 2, 2, 2, 3, 5,…
$ EX01_03                  <ord> 5, 1, 3, 1, NA, 2, NA, 2, 2, 1, 1, 3, 5, 3,
$ EX01_04                  <ord> 5, 1, 1, 1, 1, 2, 4, 1, 4, NA, 4, 3, 4, 5,…
$ STARTED                  <dttm> 2025-09-26 14:11:10, 2025-09-26 15:13:01,…
$ LASTDATA                 <dttm> 2025-09-26 14:16:16, 2025-09-26 15:21:43,…
$ TIME_SUM                 <dbl> 306, 430, 369, 555, 445, 309, 745, 318, 27…
$ FINISHED                 <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, …
$ .imp                     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
$ .id                      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,…
$ total_score_imputed_items <dbl> 5.000000, 2.500000, 2.846154, 2.461538, 2.…
```

FMI analysis:

Hide

```
quality_check_adjusted <- as.mids(quality_checkt_extracted_mutated)

quality_check_fit <- with(quality_check_adjusted, lm(total_score_imputed_items ~ T
R01 + SD01 + SD02 + SD04 + SD05 + TIME_SUM))
quality_check_est <- pool(quality_check_fit)
quality_check_est$pooled
```

term	m	estimate	ubar	b	t	dfcom
<fctr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	25	3.1709380520	1.116167e-01	0	1.116167e-01	124
TR01	25	0.3803094130	2.008786e-03	0	2.008786e-03	124
SD01	25	-0.0709869155	1.009389e-02	0	1.009389e-02	124
SD02	25	0.0109380627	3.673914e-04	0	3.673914e-04	124
SD04	25	-0.0829659963	3.796710e-04	0	3.796710e-04	124
SD05	25	0.0262467493	5.554592e-03	0	5.554592e-03	124
TIME_SUM	25	-0.0001903381	4.252819e-08	0	4.252819e-08	124

7 rows | 1-7 of 11 columns

Hide

```
summary(quality_check_est, conf.int = TRUE)
```

```
# A tibble: 7 × 12
  term        estimate  std.error  statistic     df  p.value
1 (Intercept) 3.1709380520 0.3340908342  9.4924853 122.035 2.450963e-16
2 TR01        0.3803094130 0.0448194833  8.4793350 122.035 4.203514e-14
3 SD01        -0.0709869155 0.1004683346 -0.7065601 122.035 4.811868e-01
4 SD02        0.0109380627 0.0191674556  0.5706580 122.035 5.692808e-01
5 SD04        -0.0829659963 0.0194854490 -4.2579092 122.035 4.077719e-05
6 SD05        0.0262467493 0.0745291321  0.3521674 122.035 7.252200e-01
7 TIME_SUM    -0.0001903381 0.0002062236 -0.9229692 122.035 3.578445e-01
  2.5 %    97.5 %   conf.low  conf.high
1 2.5095737789 3.8322202325 2.5095737789 3.8322202325
2 -0.2913150245 0.4687623801 -0.2913150245 0.4687623801
3 -0.2699734467 0.1278996616 -0.2699734467 0.1278996616
4 -0.0270057209 0.0488881846 -0.0270057209 0.0488881846
5 -0.1215356641 -0.0443933309 -0.1215356641 -0.0443933309
6 -0.1212906873 0.1737084186 -0.1212906873 0.1737084186
7 -0.0005985772 0.0002217901 -0.0005985772 0.0002217901
```